CSM Week 11

Agenda

(and

(or

(eval

lexical vs dynamic scoping [optional scheme] }

(and #t 2)

(or 3 4)

(and)

(or)

Python

True or and 2 or and 3 or and False

Scheme

(and)

#t

(or)

#f

```
(if 1 1 (/ 1 0))
1

(if 0 (/ 1 0) 1)
Error

(and 1 #f (/ 1 0))
#f

(and 1 2 3)
3

(or #f #f 0 #f (/ 1 0))
0

(and (and) (or))
#f

(define a 4)
a

((lambda (x y) (+ a x y)) 1 2)
7

((lambda (x y z) (y x z)) 2 1 2)
1
```

λ1 λ2

((lambda (x) (x x)) (lambda (y) 4))

4

λ1(x) [p = G]

f1: λ1 [p = G]

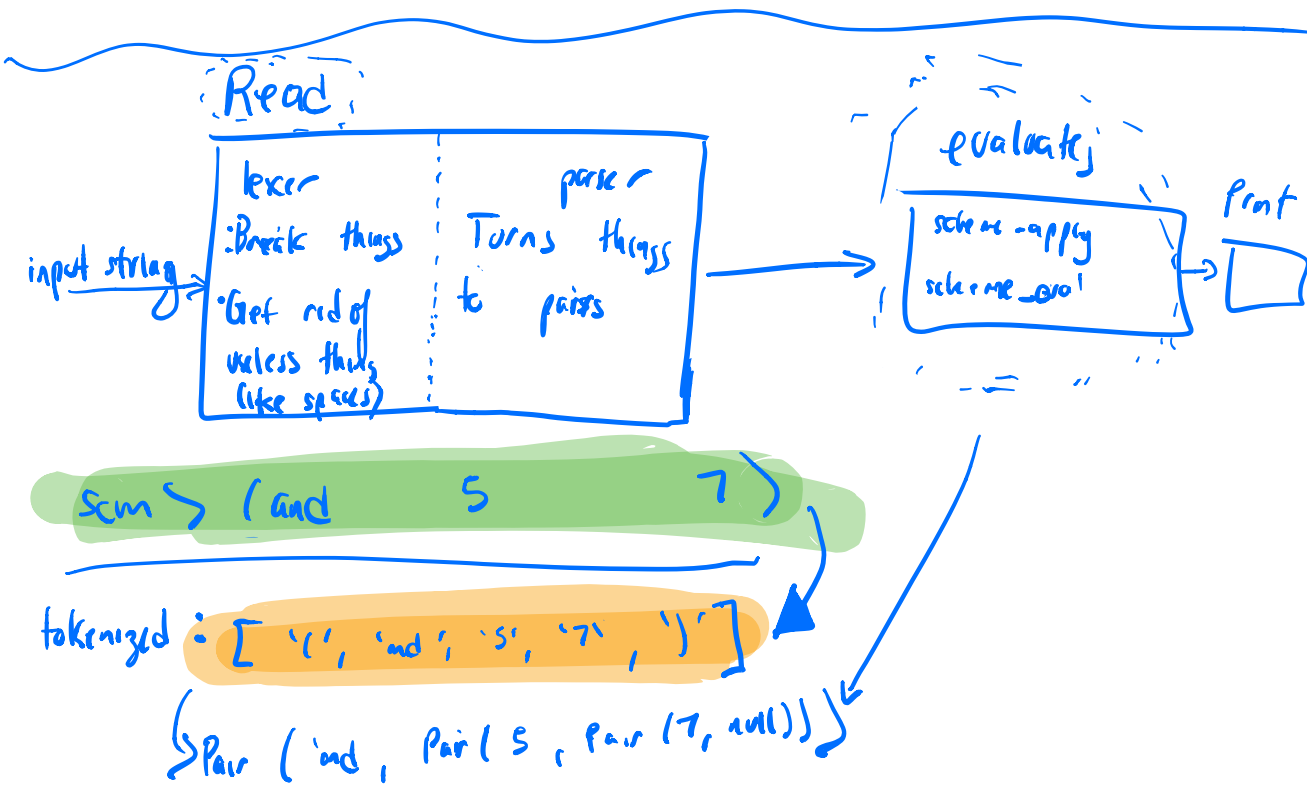
x ↳ → λ2(y) [p = G]

rv ↳ 4

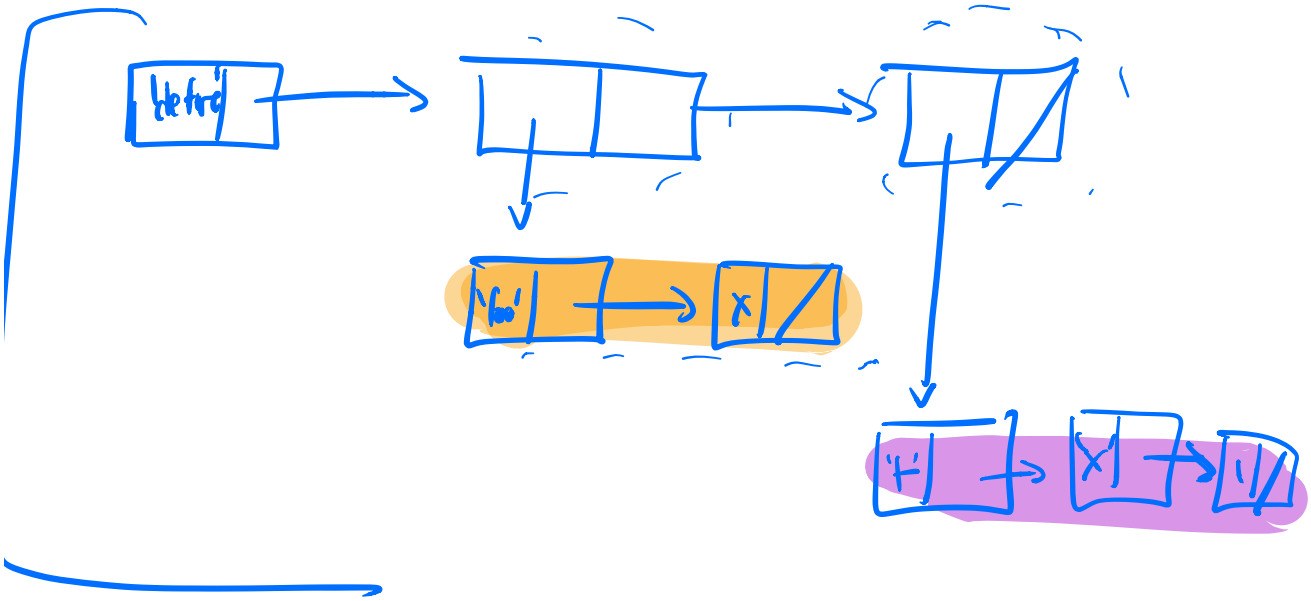
f2 λ2 [p = G]

y ↳

rv ↳ 4



scm> (define (foo x) (+ x 1))



Pair('define', Pair(Pair('foo', Pair('x', nil)),

Pair(Pair('+', Pair('x', Pair(1, nil))), nil))

```
(define (is-prefix p lst)
  (if (eq? (car lst) (car p))
      (is-prefix (cdr p) (cdr lst))
      #f
)
```

p:

'(1 2 3)

'(1)

lst:

'(1 2 3 4 5 6)

'()

```
(cond [(null? p)        #t          ]
      [(null? lst)      #f          ]
      [(eq? (car lst) (car p)) (is-prefix (cdr p)(cdr lst))]
      [else  #f ]
```